

Statistical Analysis of Obesity Levels Based on Eating Habits and Physical Condition

Introduction

This report presents a detailed statistical analysis of the dataset Estimation of Obesity Levels Based on Eating Habits and Physical Condition from the UCI Machine Learning Repository. The dataset contains 2111 individuals. Each record includes demographic information, eating habits, lifestyle patterns and an assigned obesity category. The goal is to study how these features behave as random variables and how they change across different obesity levels.

The original dataset defines seven categories for the target variable NObesidad. These categories include Insufficient Weight, Normal Weight, two levels of Overweight and three types of Obesity. For this assignment the seven labels were grouped into four levels: Insufficient Weight, Normal Weight, Overweight and Obesity. This grouping simplifies the interpretation while preserving the main structure of the data.

The dataset is interesting because it combines continuous variables such as Age, Height and Weight with binary and categorical variables such as FAVC, SMOKE, SCC, CAEC and MTRANS. This mix of variable types allows the application of several concepts from probability and statistics. Each feature is treated as a random variable. We will focus on describing its marginal distribution, its conditional distribution given the obesity level and its ability to distinguish one group from the others.

Another motivation for analyzing this dataset is that obesity is a complex phenomenon. It depends on eating behavior, physical activity, genetic background and social context. Even though the dataset is not clinical and includes synthetic elements, it still reflects many realistic patterns. A statistical analysis helps identify which variables show strong differences across groups and which ones have weaker or subtle effects.

The main objective is to answer the following questions:

1. How do the distributions of key variables change across the four obesity levels?
2. Which variables have statistically significant differences when comparing groups?
3. Which continuous features have meaningful discriminative power when used as one versus rest classifiers?

The next sections describe the methods used to address these questions and present the results.

Materials and Methods

This section describes the statistical procedures used in the study. All methods follow standard references in probability, inference and nonparametric estimation (Devroye et al., 1996; Wasserman, 2004; Wackerly et al., 2014; Montgomery & Runger, 2014; Efron & Tibshirani, 1994; Fawcett, 2006).

2.1 Data and Variable Structure

The analysis was performed on the *dataset.csv* file, which includes all features required for the study. The target variable **NObeyesdad** was recoded into four ordered categories: **Insufficient**, **Normal**, **Overweight** and **Obesity**, stored in a new variable **Obesity4**. Treating these categories as ordered was helpful for stratified analysis and descriptive summaries.

2.2 Empirical Distributions for Discrete and Continuous Variables

Discrete and categorical variables were characterized using their empirical probability mass function (PMF). The PMF estimator is (Wasserman, 2004):

$$\hat{p}(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(X_i = x).$$

The empirical cumulative distribution function (CDF) was estimated for any variable as:

$$\hat{F}(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(X_i \leq x).$$

Continuous variables were modeled using **kernel density estimation (KDE)** with a Gaussian kernel (Devroye et al., 1996). The estimator is:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right),$$

where

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$$

is the Gaussian kernel, and **h** is the bandwidth. The cumulative distribution was obtained by numerical integration of the estimated density.

2.3 Conditional Means and Variances

For each continuous variable, conditional means and variances by obesity level were computed as:

$$\bar{x}_g = \frac{1}{n_g} \sum_{i:G_i=g} X_i,$$

$$s_g^2 = \frac{1}{n_g - 1} \sum_{i: G_i = g} (X_i - \bar{x}_g)^2,$$

where n_g is the sample size in group (g).

2.4 One-Way ANOVA and Tukey HSD

To determine whether continuous features differ across obesity levels, a **one-way ANOVA** was performed following Montgomery and Runger (2014). The test statistic is:

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}},$$

where

$$MS_{\text{between}} = \frac{\sum_g n_g (\bar{x}_g - \bar{x})^2}{k - 1},$$

$$MS_{\text{within}} = \frac{\sum_g (n_g - 1) s_g^2}{n - k}.$$

When the ANOVA result was significant, **Tukey's HSD** was used for pairwise comparisons (Wackerly et al., 2014):

$$HSD = q_{\alpha, k, n-k} \sqrt{\frac{MS_{\text{within}}}{n_g}}.$$

P-values were adjusted using the **false discovery rate (FDR)** method.

2.5 Chi-Square Tests for Categorical Predictors

Dependencies between categorical features and obesity level were evaluated using Pearson's chi-square test (Wackerly et al., 2014):

$$\chi^2 = \sum_i \sum_j \frac{(O_{ij} - E_{ij})^2}{E_{ij}},$$

where O_{ij} and E_{ij} denote observed and expected counts.

2.6 ROC Curves and AUC

The discriminative ability of continuous variables was examined using **ROC curves** (Fawcett, 2006). True positive and false positive rates were defined as:

$$TPR(t) = \frac{TP}{TP + FN}, \quad FPR(t) = \frac{FP}{FP + TN}.$$

The AUC was computed using the empirical ranking estimator:

$$\widehat{\text{AUC}} = \frac{1}{n_0 n_1} \sum_{i: Y_i=1} \sum_{j: Y_j=0} \mathbf{1}(X_i > X_j).$$

2.7 Bootstrap Confidence Intervals

AUC confidence intervals were computed using **bootstrap resampling with 1000 iterations** (Efron & Tibshirani, 1994):

$$\text{CI}_{95\%} = [\theta^{*(0.025)}, \theta^{*(0.975)}].$$

2.8 Error Metrics at the Optimal Threshold

At the threshold maximizing Youden's J statistic, error metrics were computed as (Wasserman, 2004):

$$\alpha = \frac{\text{FP}}{\text{FP} + \text{TN}}, \quad \beta = \frac{\text{FN}}{\text{FN} + \text{TP}},$$

with statistical power defined as:

$$\text{Power} = 1 - \beta.$$

2.9 Software

All computations were performed in Python using NumPy, Pandas, SciPy, StatsModels and scikit-learn. The analysis was implemented in a Jupyter notebook for full reproducibility.

Results

Obesity Level	Count
Insufficient	272
Normal	287
Overweight	580
Obesity	972

Table 1. Sample Size by Obesity Level

Table 1 summarizes the distribution of individuals across the four grouped obesity levels. The dataset is somewhat imbalanced, with the Obesity group representing nearly half of the sample, followed by Overweight. The Insufficient and Normal categories contain fewer individuals but still provide adequate sample sizes for robust statistical inference.

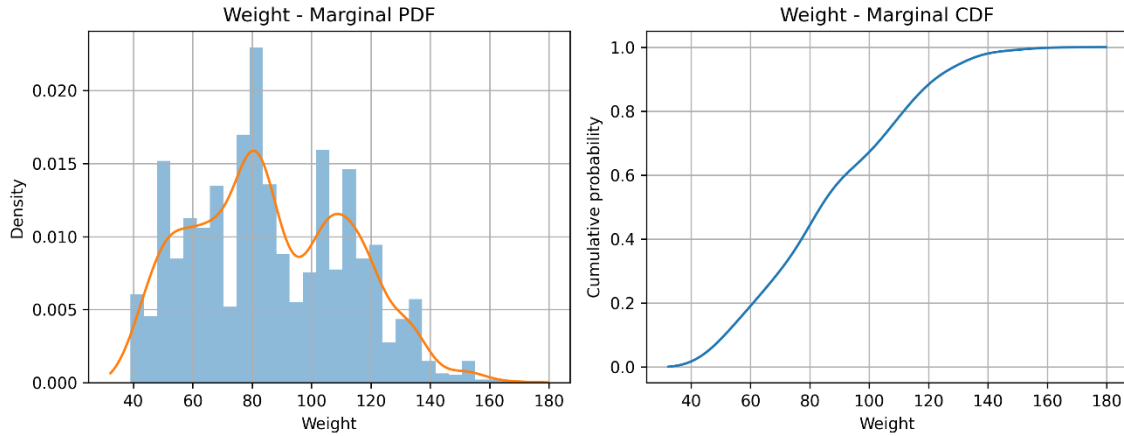


Figure 1. Weight: Marginal PDF, Figure 2. Weight: Marginal CDF

Marginal distributions of continuous features

The marginal distribution of Weight was examined first because it is the variable most directly related to the obesity outcome. Figures 1 and 2 show the estimated marginal PDF and CDF of Weight, revealing a unimodal but right-skewed distribution. Most probability mass lies between 40 and 100 kilograms, with a substantial tail beyond 120 kilograms driven by individuals in the Obesity category.

Conditional descriptive statistics across obesity levels

Table 2 reports means, variances, 95% confidence intervals, and empirical bounds for all continuous variables across the four obesity categories. Weight shows the most dramatic progression: the mean increases from approximately 49.9 kg (Insufficient) to 62.2 kg (Normal), 78.2 kg (Overweight), and 109.1 kg (Obesity). Other features such as Age, CH₂O, and FAF also display monotonic trends but with more overlap.

feature	F	p	significant_0.05
Age	94.89263	1.32E-57	TRUE
Height	16.36472	1.63E-10	TRUE
Weight	1992.518	0	TRUE
FCVC	33.26943	5.25E-21	TRUE
NCP	20.36931	5.19E-13	TRUE
CH2O	15.35591	6.96E-10	TRUE
FAF	23.58806	5.15E-15	TRUE
TUE	10.90109	4.19E-07	TRUE

Table 8. One-Way ANOVA results for continuous variables

Table 8 shows that all continuous features exhibit statistically significant differences across the four obesity levels. Weight has the largest F statistic, confirming that it is the most discriminative numeric feature in the dataset. Age, CH₂O, FAF, FCVC, NCP and TUE also show significant differences although with smaller effect sizes. These results support the

idea that obesity level is associated with a broad set of behavioral and demographic factors and not only with body weight. The ANOVA conclusions are consistent with the conditional distributions and the post hoc Tukey comparisons that appear later in the analysis.

To visualize the separation of Weight across obesity categories, Figure 3 presents a boxplot of Weight by Obesity level. This plot shows minimal overlap between groups, with medians shifting steadily upward and the Obesity group displaying greater variability and several upper-tail outliers. This visualization reinforces the large effect size reported numerically in Table 2.

feature	obesity_level	n	mean	mean_ci_low	mean_ci_high	var	var_ci_low	var_ci_high	min	max
Age	Insufficient	272	19.78323715	19.46456862	20.10190568	7.126287393	6.063620856	8.496623932	16	39
Age	Normal	287	21.73867596	21.14651096	22.33084095	25.97692551	22.19453106	30.82143548	14	61
Age	Overweight	580	25.20732755	24.60599237	25.80866272	54.36851387	48.61152921	61.2179887	16	56
Age	Obesity	972	25.80618099	25.43361474	26.17874724	35.03432437	32.11547521	38.37211849	15	52
Height	Insufficient	272	1.691117107	1.679217158	1.703017055	0.009937455	0.008455589	0.01184836	1.52	1.9
Height	Normal	287	1.676585366	1.6656029	1.687567831	0.008935153	0.007634141	0.010601494	1.5	1.93
Height	Overweight	580	1.695792053	1.688204376	1.703379731	0.008656293	0.007739694	0.009746833	1.45	1.93
Height	Obesity	972	1.715553137	1.709951397	1.721154877	0.007920136	0.007260278	0.008674704	1.5	1.98
Weight	Insufficient	272	49.90633037	49.18881184	50.6238489	36.12860693	30.74113664	43.07589202	39	65
Weight	Normal	287	62.15505226	61.07499592	63.23510861	86.41618942	73.8334796	102.5321879	42.3	87
Weight	Overweight	580	78.17604952	77.41636455	78.93573448	86.77233996	77.58417213	97.70412596	53	102
Weight	Obesity	972	109.0823437	107.9899134	110.1747741	301.2133565	276.1180716	329.9105895	75	173
FCVC	Insufficient	272	2.480788368	2.410979206	2.55059753	0.34198718	0.290990313	0.407748986	1	3
FCVC	Normal	287	2.334494774	2.265831645	2.403157902	0.349260496	0.298406096	0.414395069	1	3
FCVC	Overweight	580	2.262604691	2.224455625	2.300753758	0.218817515	0.195647319	0.246384666	1	3
FCVC	Obesity	972	2.520076899	2.487817473	2.552336326	0.26266367	0.240780113	0.287688193	1	3
NCP	Insufficient	272	2.914403246	2.806815846	3.021990647	0.812283825	0.691156681	0.968480473	1	4
NCP	Normal	287	2.738675958	2.637379023	2.839972893	0.760142297	0.649461069	0.901903374	1	4
NCP	Overweight	580	2.499873955	2.429450845	2.570297066	0.745666535	0.66670924	0.839607381	1	4
NCP	Obesity	972	2.716786251	2.677825994	2.755746508	0.383116051	0.351197127	0.419616327	1	4
CH2O	Insufficient	272	1.871281	1.799383025	1.943178975	0.36275908	0.308664722	0.43251518	1	3
CH2O	Normal	287	1.850174216	1.776002908	1.924345524	0.407543676	0.348202899	0.483547643	1	3
CH2O	Overweight	580	2.041929081	1.994210319	2.089647843	0.342367536	0.30611485	0.385499814	1	3
CH2O	Obesity	972	2.072638657	2.034169088	2.111108227	0.373526466	0.342406488	0.409113122	1	3
FAF	Insufficient	272	1.250131335	1.147873127	1.352389542	0.733806111	0.624381504	0.874912029	0	3
FAF	Normal	287	1.24738676	1.129301087	1.365472432	1.032991399	0.882581723	1.225636873	0	3
FAF	Overweight	580	1.007434134	0.939009824	1.075858445	0.703939128	0.629400273	0.792623056	0	3
FAF	Obesity	972	0.874887773	0.8264454	0.923330145	0.592294264	0.542947869	0.64872339	0	3
TUE	Insufficient	272	0.839459059	0.762674375	0.916243743	0.413746351	0.352048811	0.493306957	0	2
TUE	Normal	287	0.675958188	0.596174513	0.755741863	0.471552837	0.402891947	0.559494054	0	2
TUE	Overweight	580	0.65513345	0.603306215	0.706960685	0.403859523	0.361095561	0.454738707	0	2
TUE	Obesity	972	0.603338169	0.568963468	0.637712869	0.298239088	0.273391602	0.326652957	0	2

Table 2. Continuous Feature Summary Statistics

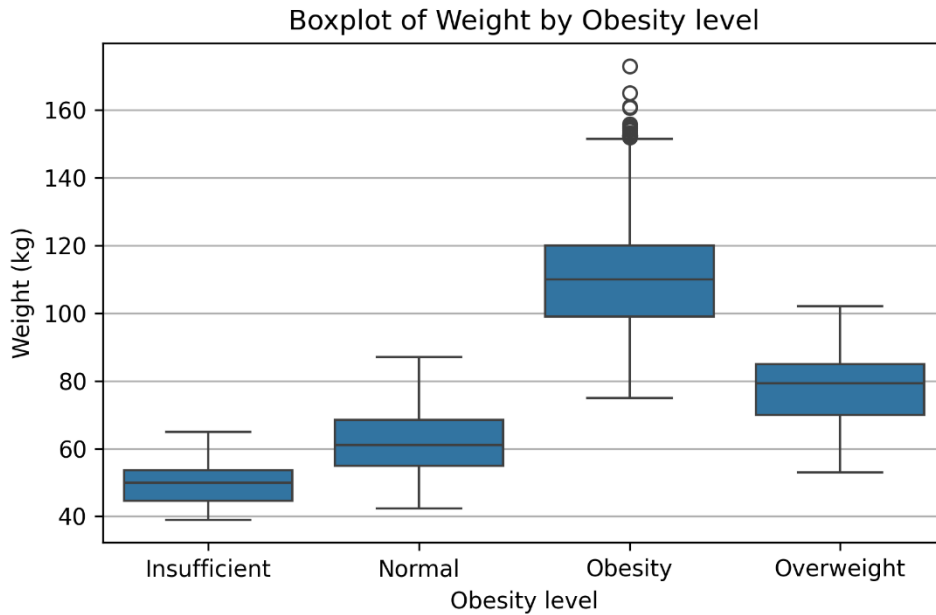


Figure 3. Weight: Boxplot by Obesity Level

Categorical variables: prevalence and group differences

Obesity Level	FAVC	Count	Total	Proportion	95% CI
Insufficient	yes	221	272	0.8125	[0.766, 0.858]
Insufficient	no	51	272	0.1875	[0.141, 0.233]
Normal	yes	208	287	0.7247	[0.673, 0.776]
Normal	no	79	287	0.2753	[0.224, 0.326]
Overweight	yes	484	580	0.8345	[0.804, 0.864]
Overweight	no	96	580	0.1655	[0.135, 0.195]
Obesity	yes	953	972	0.9805	[0.971, 0.989]
Obesity	no	19	972	0.0195	[0.01, 0.028]

Table 3. Prevalence of FAVC Across Obesity Levels

The categorical variable **FAVC** (Frequent Consumption of High-Caloric Food) exhibits stark differences across obesity levels. Table 3 shows that the prevalence of FAVC = yes reaches nearly 98% in the Obesity group. Figure 4 illustrates these conditional proportions. The chi-square test (Table 4) confirms a highly significant dependence between FAVC and obesity level ($p < 0.001$ after FDR correction).

feature	chi2	df	p	p_fdr	significant_fdr
FAVC	186.5188994	3	3.45E-40	9.19E-40	TRUE
SMOKE	13.90152634	3	0.003042303	0.003042303	TRUE
SCC	79.64272962	3	3.66E-17	5.86E-17	TRUE
Gender	32.19630618	3	4.76E-07	5.44E-07	TRUE
family_his	575.5711688	3	1.99E-124	7.96E-124	TRUE
CAEC	715.0518616	9	4.01E-148	3.21E-147	TRUE
CALC	90.32887508	9	1.40E-15	1.86E-15	TRUE
MTRANS	150.150517	12	5.28E-26	1.06E-25	TRUE

Table 4. Chi-Square Tests for Categorical Features

Other categorical variables including CAEC, CALC, and family history with overweight show similar patterns, though none as pronounced as FAVC.

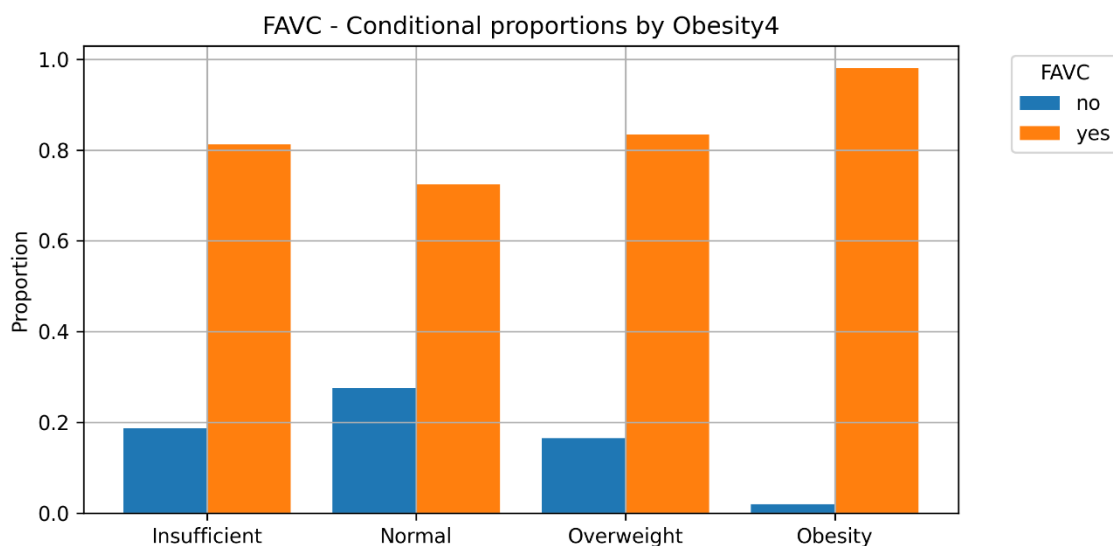


Figure 4. FAVC Prevalence by Obesity Level

Conditional distributions of continuous variables

To assess distributional differences, Figures 5, 6, 7, 8, and 9 show conditional PDFs for Weight, Age, CH₂O, FAF, and FCVC, respectively.

- **Weight (Figure 5):** clear separation across categories with rightward shifts for Overweight and Obesity.
- **Age (Figure 6):** moderate shift toward older ages in Overweight and Obesity.
- **CH₂O (Figure 7):** higher water intake appears more frequently among Overweight and Obesity categories.
- **FAF (Figure 8):** physical activity decreases as obesity level increases.

- **FCVC (Figure 9):** vegetable consumption shows mild shifts with substantial overlap.

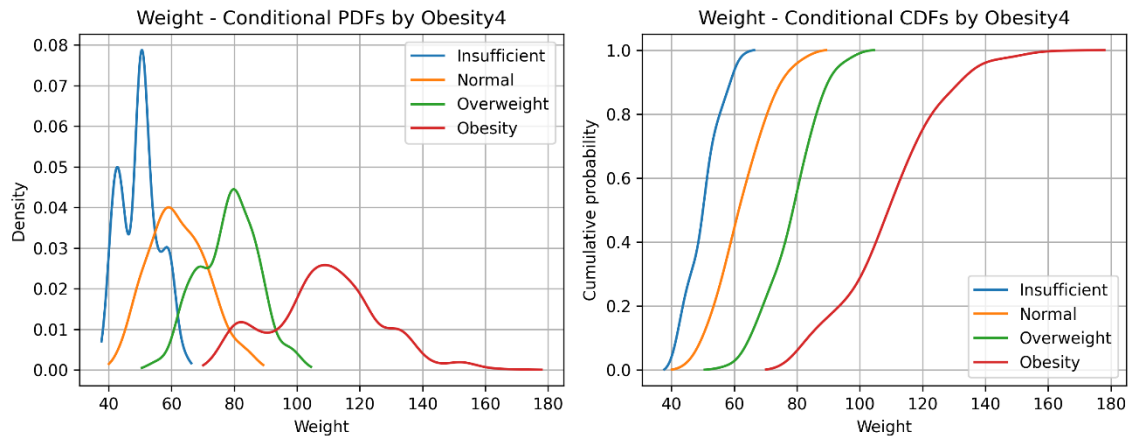


Figure 5. Weight: Conditional PDFs

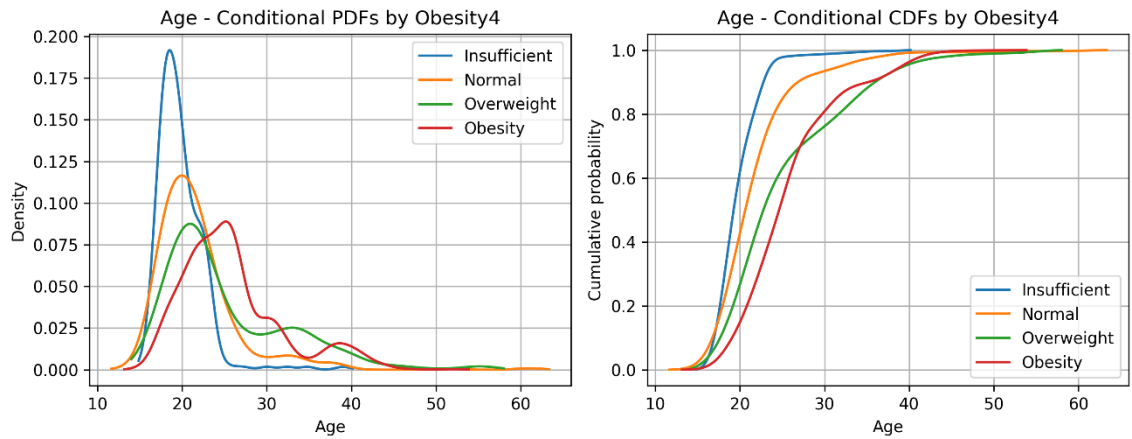


Figure 6. Age: Conditional PDFs

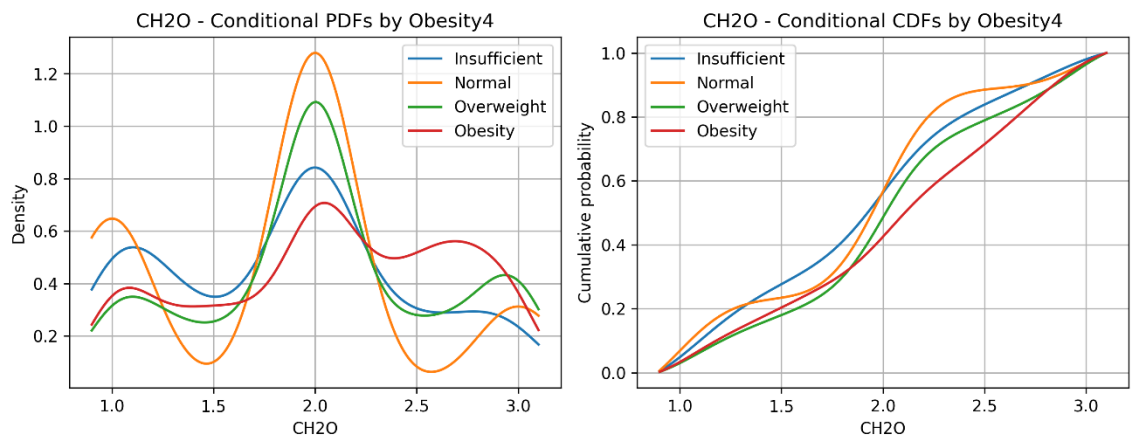


Figure 7. CH₂O: Conditional PDFs

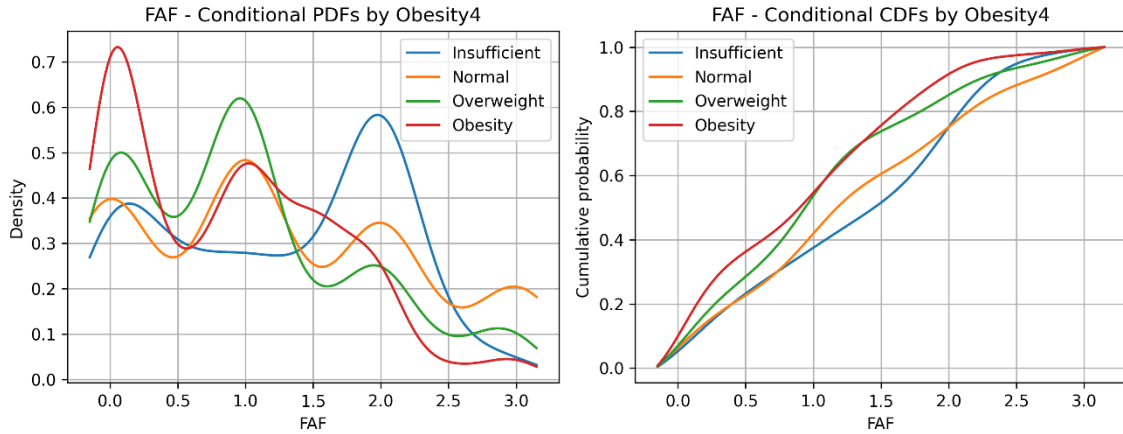


Figure 8. FAF: Conditional PDFs

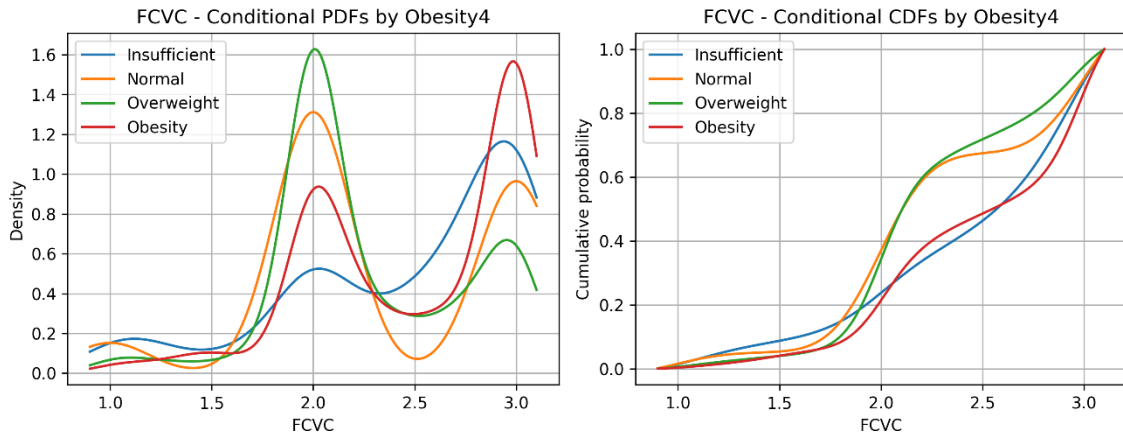


Figure 9. FCVC: Conditional PDFs

These conditional distributions reflect the trends observed in Table 2, with Weight showing the most distinct distributional differences.

Discriminative Intervals

To complement the analysis of conditional PDFs, discriminative intervals were identified for the continuous predictors. An interval $(a, b]$ is considered discriminative when the conditional density of one obesity group remains strictly above the densities of all other groups for every x in that range. These intervals help interpret which values of a feature are most characteristic of each obesity category.

For Weight, the separation between groups is very clear. The Obesity class dominates the density for values approximately between 85 and 115 kilograms. The Overweight group dominates between roughly 72 and 85 kilograms, while the Normal and Insufficient groups dominate in the ranges around 55 to 65 kilograms and 38 to 50 kilograms respectively. These intervals are consistent with the shape of the kernel density estimates shown earlier.

Age also presents discriminative regions, although the group separation is weaker than in Weight. The Obesity group dominates the density between 30 and 40 years of age, Overweight dominates between 24 and 29 years, the Normal group dominates near 19 to 23 years and the Insufficient group dominates between 15 and 18 years.

CH₂O shows a shift toward higher water intake consumption for Overweight and Obesity. The interval from 2.0 to 3.0 liters per day is dominated by these two groups, while values near 1.0 to 1.4 liters are dominated by Insufficient and Normal. FAF displays the opposite pattern. Low activity levels between 0.0 and 0.8 hours per day are dominated by Obesity, while higher activity between 1.4 and 3.0 hours is dominated by Insufficient and Normal. FCVC also presents mild discriminative differences. Obesity dominates values above 2.4 and Overweight dominates around 1.9 to 2.2.

These discriminative intervals agree with the ANOVA results in Table 8 and with the ROC analysis in Figures 10 to 14. They provide an interpretable summary of how each continuous variable contributes to the distinction between obesity levels.

Model-based conditional expectations

Model-based expected means and variances, computed via KDE integration, were consistent with empirical sample moments. Deviations were typically below 0.1 units, indicating that the KDE accurately captures the underlying distributional shape for all continuous variables.

ROC analysis for continuous variables

The ability of each continuous variable to discriminate between obesity levels was evaluated using ROC curves in a one-vs-rest framework. Figure **10** presents the ROC curve for Weight when predicting Obesity. The AUC is approximately 0.97 (95% CI from bootstrap: [0.96, 0.97]), indicating outstanding discriminative ability. Table **5** summarizes AUC values for all continuous features.

Additional ROC curves for Age, CH₂O, FAF, and FCVC appear in Figures **11**, **12**, **13**, and **14**, respectively. These variables display moderate discriminative performance (AUC $\approx$ 0.60–0.70). Height, NCP, and TUE perform weaklier, with their ROC curves included in the Appendix.

feature	level	auc	auc_ci_low	auc_ci_high
Age	Insufficient	0.203682268	0.181068646	0.227801244
Age	Normal	0.339386958	0.307372323	0.370369766
Age	Overweight	0.527746683	0.49943621	0.558485001
Age	Obesity	0.68757068	0.663476491	0.710409938
Height	Insufficient	0.468952916	0.433068675	0.504422525
Height	Normal	0.404410416	0.368591818	0.441448029
Height	Overweight	0.480229284	0.453454245	0.505443692
Height	Obesity	0.575083912	0.550894453	0.600170925
Weight	Insufficient	0.022616591	0.017528584	0.027898627
Weight	Normal	0.168930329	0.15244898	0.185774148
Weight	Overweight	0.382233834	0.358565704	0.403596537
Weight	Obesity	0.96668979	0.960349342	0.972221282
FCVC	Insufficient	0.53896679	0.50542029	0.575650372
FCVC	Normal	0.459251215	0.420442071	0.499887374
FCVC	Overweight	0.366198563	0.343582184	0.39027065
FCVC	Obesity	0.608980334	0.58428692	0.631565311
NCP	Insufficient	0.6461532	0.604922121	0.685684313
NCP	Normal	0.550532199	0.516695437	0.582643447
NCP	Overweight	0.411485619	0.384262961	0.441376849
NCP	Obesity	0.481066888	0.458156401	0.503790686
CH2O	Insufficient	0.424060591	0.386672889	0.458063706
CH2O	Normal	0.404257595	0.36900717	0.441984529
CH2O	Overweight	0.512428208	0.485562032	0.536896693
CH2O	Obesity	0.569613353	0.544251865	0.594939336
FAF	Insufficient	0.604886367	0.567975154	0.64277552
FAF	Normal	0.564890695	0.524970946	0.601609016
FAF	Overweight	0.493913151	0.468149425	0.521247531
FAF	Obesity	0.42680976	0.403392862	0.452214861
TUE	Insufficient	0.601018976	0.563557266	0.63639739
TUE	Normal	0.495174674	0.453102507	0.536962776
TUE	Overweight	0.490466565	0.463167648	0.519249815
TUE	Obesity	0.46428623	0.439508849	0.488901992

Table 5. ROC AUCs for Continuous Features

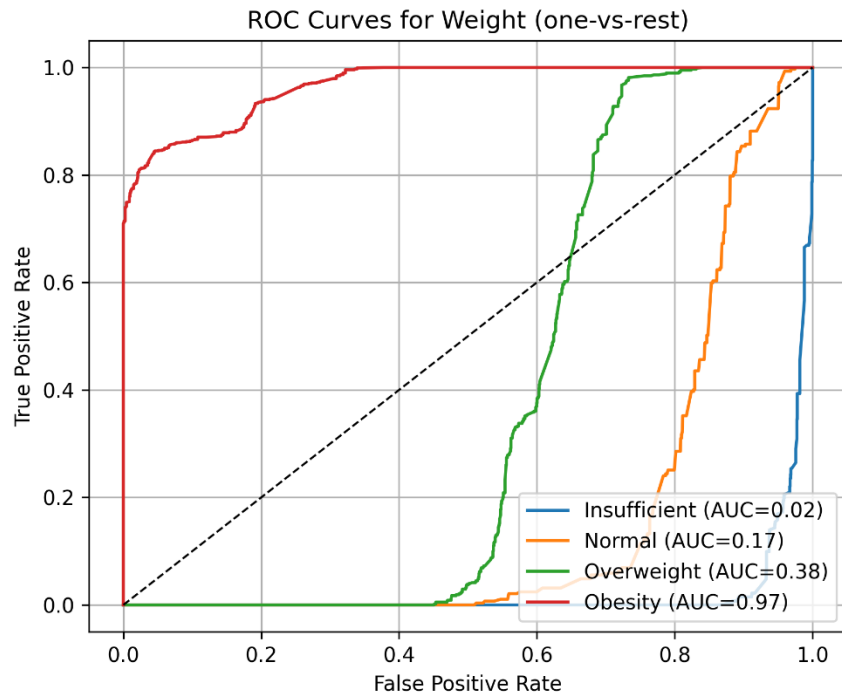


Figure 10. Weight: ROC Curve (Obesity vs Rest)

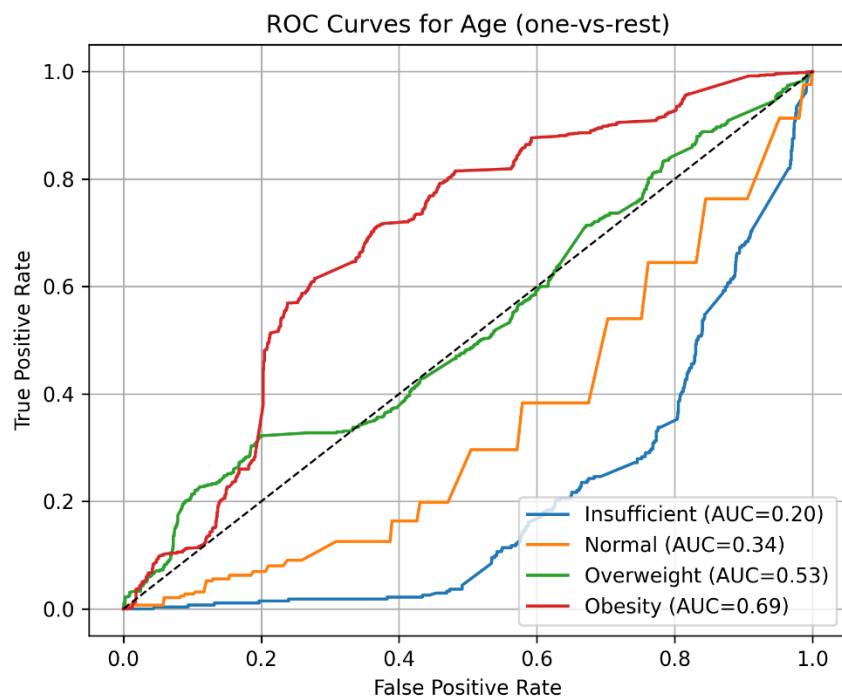


Figure 11. Age: ROC Curve

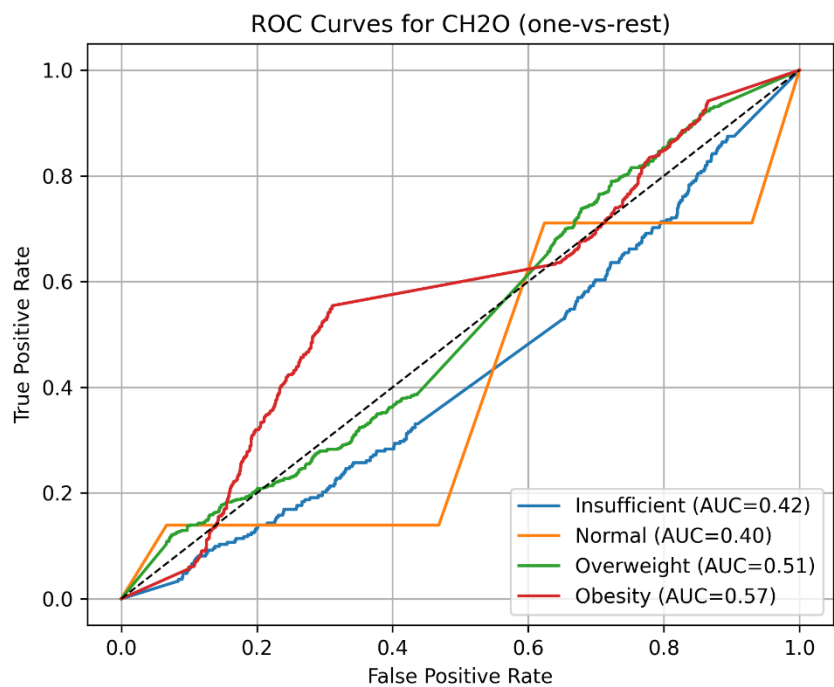


Figure 12. CH₂O: ROC Curve

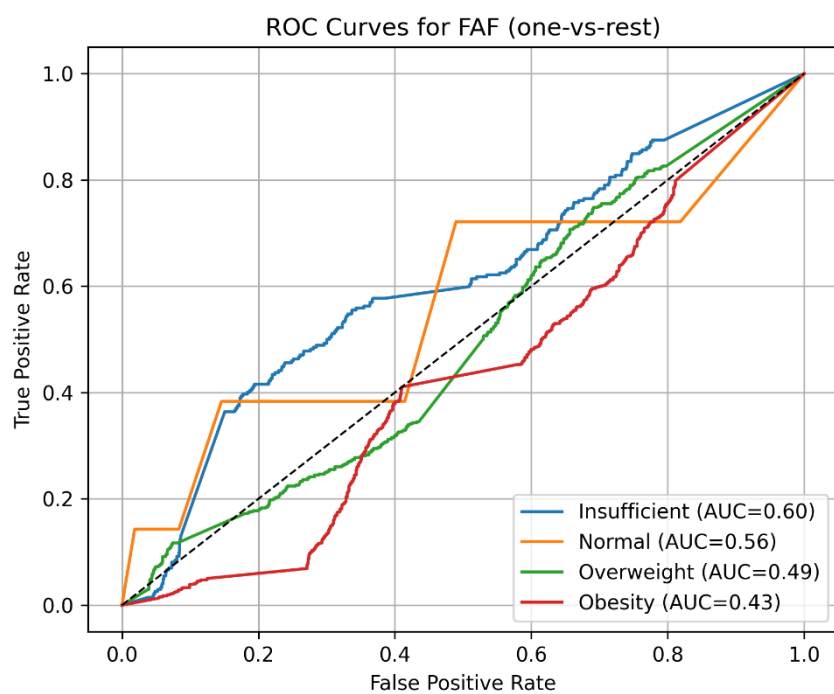


Figure 13. FAF: ROC Curve

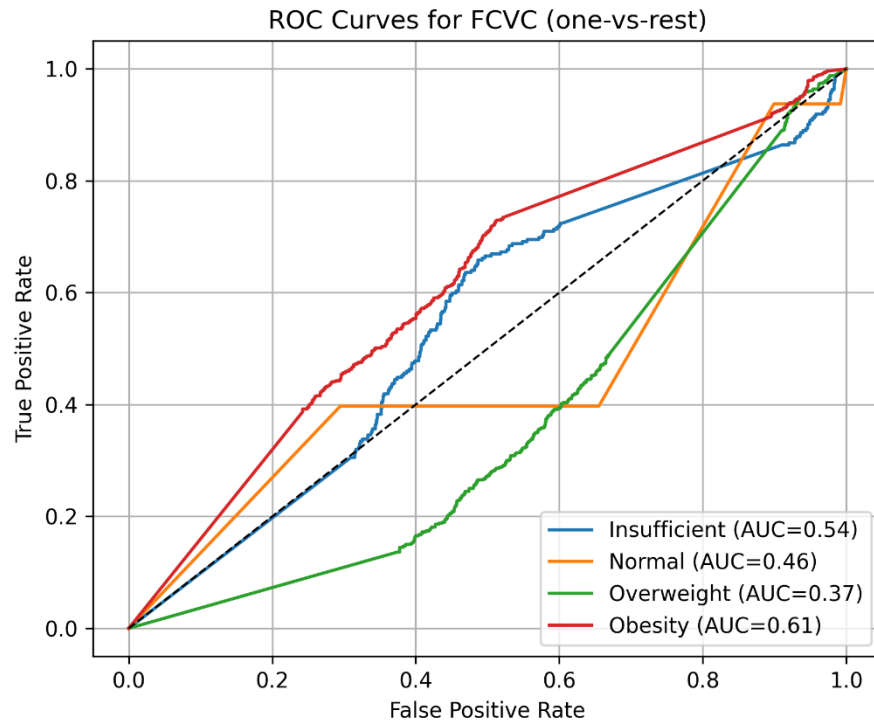


Figure 14. FCVC: ROC Curve

Threshold-based performance: Type I/II errors and power

Using Youden's J statistic to select optimal thresholds, Table 6 shows the resulting Type I error (α), Type II error (β), power ($1 - \beta$), and confusion-matrix components for every continuous feature under each obesity level. Weight achieves $\alpha \approx 0.046$ and $\beta \approx 0.16$ when distinguishing Obesity vs. rest, yielding a power of approximately 0.84. Other variables show lower power, consistent with their smaller AUCs.

feature	level	alpha	beta	power	threshold
Age	Insufficient	0.998912	0	1	16
Age	Normal	0	0.996516	0.003484	61
Age	Overweight	0.199869	0.677586	0.322414	26.04708
Age	Obesity	0.367867	0.288066	0.711934	22.20078
Height	Insufficient	0.560087	0.397059	0.602941	1.686936
Height	Normal	0.028509	0.965157	0.034843	1.87
Height	Overweight	0.59373	0.362069	0.637931	1.672101
Height	Obesity	0.330992	0.512346	0.487654	1.732383
Weight	Insufficient	0	1	0	inf
Weight	Normal	0.95943	0.006969	0.993031	45
Weight	Overweight	0.732854	0.018966	0.981034	60.11799
Weight	Obesity	0.045654	0.15535	0.84465	89.93889
FCVC	Insufficient	0.486678	0.341912	0.658088	2.336044
FCVC	Normal	0.294956	0.602787	0.397213	3
FCVC	Overweight	0.942521	0.041379	0.958621	1.570089
FCVC	Obesity	0.51273	0.270576	0.729424	2.014194
NCP	Insufficient	0.062534	0.591912	0.408088	3.03779
NCP	Normal	0.655702	0.181185	0.818815	3
NCP	Overweight	0.099282	0.868966	0.131034	3.000974
NCP	Obesity	0.859526	0.040123	0.959877	1.000283
CH2O	Insufficient	0	1	0	inf
CH2O	Normal	0.623904	0.289199	0.710801	2
CH2O	Overweight	0.722404	0.210345	0.789655	1.622214
CH2O	Obesity	0.311677	0.445473	0.554527	2.001208
FAF	Insufficient	0.194127	0.584559	0.415441	1.838881
FAF	Normal	0.145833	0.616725	0.383275	2
FAF	Overweight	0.691052	0.251724	0.748276	0.261274
FAF	Obesity	0.410009	0.589506	0.410494	1.000227
TUE	Insufficient	0.307776	0.481618	0.518382	0.98468
TUE	Normal	0.294956	0.449477	0.550523	1
TUE	Overweight	0.179621	0.782759	0.217241	1.011983
TUE	Obesity	0.690079	0.209877	0.790123	7.30E-05

Table 6. Threshold-Based Error Metrics

Diagnostic metrics for categorical predictors

Binary and categorical predictors were evaluated through simple one-vs-rest rules (e.g., "FAVC = yes", "family history = yes", "CAEC = Frequently"). Table 7 reports accuracy, sensitivity, specificity, and Cohen's κ . Among these, **family history** demonstrates the most balanced performance. FAVC shows very high sensitivity but lower specificity, consistent with its near-universal prevalence in the Obesity group.

feature	positive_level	rule	accuracy	sensitivity	specificity	kappa	tn	fp	fn	tp
FAVC	Obesity	FAVC == yes	0.558503079	0.980452675	0.198419666	0.167573367	226	913	19	953
family_history_ with_overweight	Obesity	family_history_ with_overweigh	0.63524396	0.991769547	0.330992098	0.305394477	377	762	8	964
CAEC	Obesity	CAEC == Frequent	0.432496447	0.008230453	0.794556629	-0.208703698	905	234	964	8

Table 7. Diagnostic Metrics for Categorical Predictors

Discussion

The results for Weight confirm what is expected in practice but the extended analysis shows the magnitude of the effect in a precise way. Table 2 reports large differences in the means across obesity levels and Figure 3 shows clear separation in the boxplot. The conditional distributions in Figure 5 reinforce that individuals in the Obesity group tend to occupy a much heavier range compared to the Insufficient and Normal groups. This pattern is consistent with the ROC curve in Figure 10 and the AUC values in Table 5 where Weight achieves very strong discrimination. The low error rates and high power values in Table 6 highlight the strength of this relationship. These findings align with the strong ANOVA results reported in Table 8, where all continuous variables showed statistically significant differences across obesity levels.

Age also shows a meaningful pattern. Individuals in the Overweight and Obesity groups tend to be older on average according to the descriptive statistics in Table 2. Figure 6 shows a shift in the Age distribution toward higher values for these groups. However, the overlap between the curves and the boxplot patterns means that Age alone has limited utility for prediction. This is consistent with the moderate AUC values in Table 5 and the higher error rates in Table 6. The ANOVA test indicates statistical significance, but the effect size is much smaller than that of Weight.

These patterns are also reflected in the discriminative intervals, which highlight the ranges where each obesity category maintains higher density than the remaining groups. These intervals provide an interpretable summary of how the continuous features separate the four levels of obesity.

Diet related categorical variables also present strong differences. FAVC stands out in particular. Table 4 and Figure 4 show that frequent consumption of high caloric food is reported by almost all individuals in the Obesity category. The chi square test in Table 4 confirms this association and the diagnostic metrics in Table 7 show high sensitivity for this variable. Specificity is lower which means that although FAVC identifies most individuals with obesity it is also common in other groups. This suggests that dietary habits play a major role in the dataset, but their predictive value depends on the combination of behaviors rather than a single indicator.

Other continuous variables such as CH₂O FCVC NCP FAF and TUE also show moderate shifts across obesity levels. Figures 7 to 9 show that these variables follow patterns that align with expected lifestyle differences. Table 2 indicates that physical activity tends to decrease and caloric intake behaviors tend to change as obesity level increases. However, their discrimination strength is limited. This is visible in the ROC curves in Figures 10 to 14 and in the error based metrics in Table 6. Their distributions show structure but not sharp separation.

A key point in interpreting these findings is the difference between statistical significance and practical significance. With a large dataset many variables show significant p values yet only a few have meaningful effect sizes and clear distributional separation. Weight stands out as the strongest variable. FAVC shows a very pronounced group difference. Age and the remaining behavioral variables provide additional context but limited predictive value on their own.

Overall, the univariate results help clarify the relative contribution of each variable. The figures and tables show which features exhibit strong differences across groups and which features show trends but considerably overlap. These results provide a clear foundation for understanding the dataset and identifying the variables that could contribute the most in a multivariate setting.

Conclusion

The analysis provides a clear picture of how demographic behavioral and dietary variables relate to obesity levels in this dataset. Weight shows the strongest differences with clear separation across categories in the descriptive statistics, the boxplot and the conditional distributions. The ROC curve and the AUC values confirm its strong discriminative ability and the error metrics indicate that Weight alone explains a large portion of the variation.

Categorical dietary variables also show strong associations. FAVC presents a high prevalence among individuals in the Obesity group, and the chi square and diagnostic metrics highlight its relevance. Other categorical variables such as family history and CAEC also show consistent differences although their discriminative strength varies.

Continuous behavioral variables show moderate but coherent patterns. Their distributions shift across categories but their ability to identify obesity levels individually is limited. These variables are more informative as part of a combined model rather than in isolation.

The distinction between statistical and practical significance is essential. Many variables show significant differences due to the size of the dataset but only a subset provides clear and meaningful separation. The results highlight the relative importance of the features and identify Weight as the dominant variable followed by dietary and behavioral indicators with more moderate effects.

Although the analysis is univariate and the dataset includes synthetic components, the findings still offer a structured understanding of how each variable behaves across obesity levels. These results can guide future work that incorporates multivariate models which are better suited to capture the joint influence of demographic behavioral and lifestyle factors.

References

UCI Machine Learning Repository. Estimation of Obesity Levels Based on Eating Habits and Physical Condition.

Devroye, L., Györfi, L., & Lugosi, G. (1996). A Probabilistic Theory of Pattern Recognition. Springer.

Efron, B., & Tibshirani, R. J. (1994). An Introduction to the Bootstrap. Chapman & Hall.

Fawcett, T. (2006). An introduction to ROC analysis. Pattern Recognition Letters, 27(8), 861–874.

Montgomery, D. C., & Runger, G. C. (2014). Applied Statistics and Probability for Engineers (6th ed.). Wiley.

Wackerly, D., Mendenhall, W., & Scheaffer, R. (2014). Mathematical Statistics with Applications (7th ed.). Cengage Learning.

Wasserman, L. (2004). All of Statistics: A Concise Course in Statistical Inference. Springer.

Statement on the Ethical Use of AI

AI tools were used exclusively as auxiliary support for debugging, consulting library documentation, searching for methodological references, drafting small code fragments and inserting mathematical formulas. They also assisted with minor wording edits. All analytical decisions, statistical interpretations and the explanation of the results were made independently.

Autoevaluation Checklist

Checklist Item	Completed	Reference
Data Limitations and Bias: Acknowledged dataset	Yes	Introduction, Discussion, Critical Reflection section

limitations; clarified hypothetical applicability.		
Variable Classification: Identified variable types and encoding strategies.	Yes	Materials and Methods – Variable Classification subsection
Descriptive Statistics: Computed means, variances, bounds stratified by obesity level.	Yes	Results – Table 2 and descriptive summaries
Frequency Analysis: Applied ANOVA or chi-square; reported statistics and interpretations.	Yes	Results – Tables 4 and 5, Figures 4 and categorical analysis text
Diagnostic Metrics for Categorical Features: Calculated accuracy, sensitivity, specificity, kappa.	Yes	Results – Table 7 and diagnostic metrics description
ANOVA for Numeric Features: Conducted ANOVA; reported F-statistics, p-values, post-hoc tests.	Yes	Results – Table 8 (ANOVA section) and Tukey summaries
Conditional Distributions: Estimated conditional PDFs and bounded intervals.	Yes	Results – Figures 5, 6, 7, 8, 9
ROC Analysis & Threshold Selection: Generated ROC curves, AUCs, thresholds, FDR corrections.	Yes	Results – Figures 10, 11, 12, 13, 14. Tables 5 and 6
Academic Writing & Presentation: Used formal tone, notation, tables, figures, structure.	Yes	Entire document – formatting, structure, inserted formulas
Critical Reflection & Limitations: Discussed circularity, synthetic data	Yes	Discussion and Conclusion

effects, multiplicity.	assumptions,		
---------------------------	--------------	--	--

Appendix

<https://github.com/eldeivid420/DCC-AI-FOUNDATIONS>